

UMER WALI

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PROFESSIONAL SUMMARY

Full-stack developer and team lead with hands-on experience directing engineering teams from concept to production. Completed a Backend AI Engineering internship at FlyRank AI, building production-grade AI backend systems. Led a 3-developer build team on a live multi-vendor commerce platform, owning architecture, task delegation, code review, and delivery timelines under a senior architect as mentor. Background spans full-stack web development (Next.js, React, Node.js, PostgreSQL), applied AI/LLM backend engineering, client and resource coordination through freelance consulting, and systems programming, backed by a cybersecurity degree.

EXPERIENCE

Backend AI Engineer — Intern — FlyRank AI

Islamabad, Pakistan (Remote) | 2026

- **Backend AI Engineering track:** selected for FlyRank AI's Backend AI Engineering internship track, focused on server-side AI systems — API contracts, retrieval-backed answer flows, structured outputs, tool-calling workflows, and evaluation harnesses.
- **Applied AI backend work:** designing and building AI backend workflows (RAG pipelines, structured-output systems, agent/tool-calling loops) with emphasis on correctness, grounding, and failure-mode handling, reviewed against rubric-based evaluation sets.
- **Delivery & documentation:** documenting design decisions, verification notes, and recommended improvements for each workflow shipped, in line with FlyRank AI's evidence-based review process.

RELEVANT TECHNICAL SKILLS

- **Languages & Systems:** C++ (C++20, low-latency), Python, TypeScript/JavaScript, Linux Systems Programming (kernel bypass, epoll), multithreading & concurrency, memory alignment, lock-free data structures
- **Full-Stack & Web:** Next.js 14, React/Vite, Node.js, Express, Medusa.js, PostgreSQL, MySQL, Prisma, Redis, Stripe Connect
- **Agentic AI & Orchestration:** LangGraph, Multi-Agent Systems, ReAct (Reasoning + Action) Pattern, Function/Tool Calling, Human-in-the-Loop (HITL) Guardrails
- **LLMs & Multimodal AI:** Google Gemini 3.5 Flash Lite (Native SDK), Gemini Vision, Deepgram Nova-3 (STT), Retrieval-Augmented Generation (RAG)
- **Backend & APIs:** Python 3.14, FastAPI, AsyncIO, Uvicorn, REST APIs, Meta Official WhatsApp Business Platform
- **Databases & DevOps:** PostgreSQL, Async SQLAlchemy, Connection Pooling, Docker, Linux/Ubuntu VPS, Git
- **Trading & Data Systems:** Exchange/CLOB API connectivity, order execution logic, tick-to-trade latency optimization, satellite/geospatial data processing (Sentinel-2, SAR, NDVI/NDRE)
- **Cybersecurity & Networking:** Network Security, Malware Analysis, Kali Linux, Nmap, Burp Suite, NIST Frameworks, MITRE ATT&CK, Google Chronicle (SOAR)

PROJECT EXPERIENCE

Rabta AI — Enterprise Dual-Agent WhatsApp Commerce Engine — Personal Project

Backend AI Engineer / Full-Stack AI Systems | 2026

Tech Stack: Python 3.14, FastAPI, LangGraph, Google Gemini 3.5 Flash Lite (ReAct), Deepgram Nova-3, PostgreSQL, SQLAlchemy Async, Meta WhatsApp Business Platform, Docker, Linux VPS

- **Multi-Agent Orchestration (LangGraph):** architected a synchronized dual-agent state machine bifurcated into:
 - **Customer Sales Closer:** warm, culturally grounded Roman Urdu sales persona operating at temperature 0.5, driving progressive lead intake funnels (Name, City, Address, Contact) and proactive deal closures.
 - **Owner Co-Pilot:** deterministic ReAct assistant operating at temperature 0.2 for natural language catalog mutations, stock toggling, and margin controls directly through WhatsApp.
- **Tool Calling & Database RAG:** bound native Gemini function-calling tools (search_catalog, get_product_photos, check_delivery_policy, get_payment_bank_details) to async PostgreSQL queries, completely eliminating catalog hallucinations.
- **Multimodal Intelligence:**
 - **Vision:** audited studio firearm catalog photos against physical SKUs using Gemini Vision rollmark detection to eliminate weapon mismatches.
 - **Voice:** integrated Deepgram Nova-3 via async streaming WebSockets for low-latency voice note processing.
- **Edge Messaging & Concurrency:** deployed on Ubuntu Linux via Docker with asynchronous connection pooling, handling 20+ concurrent customer sessions seamlessly while maintaining natural human typing delays and presence broadcasts.

HFT Arbitrage Bot on Polymarket — Personal Project

Remote / Google Cloud (Singapore) | 2025 – Present

- **C++ Low-Latency Market Data & Execution System:** architected and deployed a C++ high-frequency arbitrage bot on the Polymarket V2 CLOB API, executing a statistical latency arbitrage and structural Dump Hedge strategy, generating \$14,200 P&L with a 68% win rate.
- Engineered sub-millisecond pipelines — native POLY_1271/ERC-7739 C++ signing, zero-allocation hot paths, and lock-free data structures with strict memory alignment — eliminating prior Python latency bridges.
- Deployed on Google Cloud (Singapore) compute-optimized instances for exchange proximity, managed entirely via CLI, achieving tick-to-trade latency < 50 μ s and 200 orders/sec burst throughput.
- Designed FOK (Fill-Or-Kill) dual-leg order execution to eliminate leg risk, and conducted a full malware analysis/security audit of a predecessor Python bot, driving a ground-up secure rewrite in C++20.

Cybersecurity Incident Management System (CIMS) — GIKI — CS-232 DBMS Course Project

Topi, KPK | 2025

- Built a full-stack incident management system on Node.js, Express, React/Vite, and PostgreSQL, with a relational schema for real-time logging and audit-trail tracking.
- Implemented secure JWT authentication, bcrypt password hashing, and RBAC middleware for multi-role access and system integrity.

From Tribe — Multi-Vendor E-Commerce Marketplace — Zero Point Intel — Switzerland / Remote

Lead Developer | 2025

- Led a 3-developer build team under mentorship of a senior cloud architect to ship a multi-vendor marketplace connecting independent EU artisan brands with customers.
- Built the Next.js 14 storefront and Medusa.js commerce backend (multi-vendor order routing, split payouts) with a PostgreSQL/Prisma data layer, Stripe Connect payouts, and Redis caching.
- Replaced ad-driven marketplace dynamics with a flat 13% commission and zero listing fees, letting small brands compete on quality and discovery instead of ad spend.

Acreon — Satellite-Based Supply-Chain Risk Monitoring — Zero Point Intel — Switzerland / Remote

2026

- Built a satellite-based risk platform for food buyers, surfacing up to 8 weeks of early lead time on crop stress, flooding, and vigor drops from Sentinel-2/Copernicus imagery.
- Processed SAR (radar) and optical satellite data to compute NDVI/NDRE vegetation indices in Python, turning raw remote-sensing data into actionable risk signals.

EDUCATION

- **BS Cybersecurity — GIKI** | Topi, KPK | 2026
- **HSSC in Pre-Engineering + Mathematics (A Grade) — Abbottabad Public School** | Abbottabad, KPK | Jun 2022

OTHER EXPERIENCE

Freelance Developer & Consultant — Various Clients

Remote | 2024 – Present

- Delivered custom software/web projects for international clients across Switzerland and remote markets, managing full development lifecycle from scoping to delivery.

CERTIFICATIONS & CREDENTIALS

- Building AI Agents and Agentic Workflows Specialization (IBM) — Sep 2026
- Google AI Essentials Specialization (Google) — Sep 2026
- Google Security Operations — SOAR Analyst (Google) — Jun 2026
- Oracle Certified AI Foundations Associate — May 2024
- Claude Code 101 Certification (Anthropic)
- Kali Linux (Coursera) — May 2024